

ANKIT KUMAR ACADEMY

ADVANCED GENERATIVE & AGENTIC AI MASTER PROGRAM

Master Generative AI, Build AI Agents, and Automate the
Future

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PROGRAM OVERVIEW

Transforming Aspirations into AI Expertise



Generative AI Fundamentals

Master core concepts and real-world applications across diverse industries.



Build AI Tools

Learn to develop custom AI-powered tools for professional automation.



AI Agent Automation

Design autonomous agents that can think, plan, and execute complex workflows.

Target Audience

Professionals

Developers

Entrepreneurs

Students

Business Analysts

Join Ankit Kumar Academy for world-class AI training.

MODULE 1: FOUNDATIONS OF AI

Understanding the Core Technologies & Evolution

AI EVOLUTION

- ▶ Traditional Machine Learning
- ▶ Deep Learning Breakthroughs
- ▶ The Rise of Generative AI
- ▶ Predictive vs. Generative Models

CORE MECHANICS

- ▶ How AI Systems Learn
- ▶ Decision-Making Processes
- ▶ Multimodal AI Integration
- ▶ Text, Image, Audio, & Video

INDUSTRY IMPACT

- ▶ AI Content Creation Tools
- ▶ Marketing & Finance Automation
- ▶ Healthcare Diagnostics
- ▶ Enterprise AI Assistants

Hands-on Experience

Real-world examples of AI text generation and content creation automation tools.

MODULE 2: AI TOOLS & PRODUCTIVITY

The AI Stack

- ▷ ChatGPT & Claude
- ▷ Gemini & Perplexity AI
- ▷ Research & Writing
- ▷ Document Generation

Business Automation

- ▷ AI Email Automation
- ▷ Document Analysis
- ▷ Report Generation
- ▷ Workflow Optimization

Developer Workflow

- ▷ Working with AI APIs
- ▷ GitHub Integration
- ▷ Managing AI Projects
- ▷ Sharing Workflows

30–
50%

Cost reduction achieved through intelligent business automation and AI-driven productivity systems.

MODULE 3: LARGE LANGUAGE MODELS

The Engine Behind the AI Revolution

Core Mechanics

■ Tokens & Tokenization

Breaking text into units for model processing.

■ Context Windows

Managing the volume of data models process at once.

■ Attention Mechanisms

Focusing on relevant parts of input for accuracy.

Popular Models

■ Proprietary Leaders

GPT (OpenAI), Claude (Anthropic), and Gemini (Google).

■ Open-Source Power

Leveraging Llama, Mistral, and community models.

■ Capabilities

Text Generation, Q&A, Code, and Reasoning.

Optimization

■ Cost Management

API cost optimization and efficient prompt design.

■ Handling Limitations

Mitigating Hallucinations, Bias, and Context Limits.

■ Hands-on Examples

Building LLM-based chatbots and Q&A systems.

MODULE 4: PROMPT ENGINEERING MASTERY

The Art of Communicating with AI

📄 Core Techniques		♟️ Advanced Strategies		✍️ Optimization	
ZERO-SHOT	Direct instructions without examples.	ROLE PROMPTING	Assigning specific expert personas.	ACCURACY	Reducing hallucinations effectively.
FEW-SHOT	Providing examples for better results.	CONTEXT	Providing deep background information.	EVALUATION	Techniques for output quality control.
CHAIN-OF-THOUGHT	Guiding step-by-step reasoning.	STRUCTURED	Using templates and frameworks.	REAL-WORLD	Coding, marketing, and content use cases.

 **Hands-on Project**

Creating a focused AI prompt library and a compact specialized assistant for your workflows.

MODULE 5: MULTIMODAL GENERATIVE AI

Beyond Text: Visuals, Audio, and Video Innovation



Visual Content

AI Art & Design Automation. Create stunning visuals from text descriptions.

Midjourney

DALL-E 3

Stable Diffusion



Audio & Voice

Natural Voice Synthesis & Narration. Automated voiceovers and audiobooks.

ElevenLabs

Murf AI

Play.ht



Video Generation

Text-to-Video & Marketing Content. Automated high-quality video production.

Runway Gen-2

Pika Labs

OpenAI Sora



Vision-Language

AI Image Analysis & Captioning. Extracting deep insights from visual data.

GPT-4V

Claude 3 Vision

Gemini Pro

MARKET GROWTH: 36.8% CAGR

PROJECTED \$10.89B BY 2030

MODULE 6: RETRIEVAL AUGMENTED GENERATION

Grounding AI in Your Own Proprietary Data

Why RAG?

- ACCURACY** 70–90% reduction in LLM hallucinations.
- REAL-TIME** Access to live, proprietary business data.
- TRUST** 95–99% time-sensitive accuracy for users.

The Pipeline

- INGESTION** Process and chunk complex documents.
- RETRIEVAL** Semantic search and ranking systems.
- GENERATION** Grounded responses with citations.

Core Tech

- EMBEDDINGS** Numerical text representations.
- VECTOR DB** Pinecone, Weaviate, Chroma, Qdrant.
- ADOPTION** Used by Siemens, Morgan Stanley, Amazon.

Market Growth

RAG market is expanding rapidly as enterprises prioritize data security.

39.6% CAGR

MODULE 7: INTRODUCTION TO AGENTIC AI

AI Systems that Think and Act Autonomously

Core Concepts

Perceive, decide, and act independently with goal-oriented behavior.

Agent Types

Task Agents, Planning Agents, and Collaborative Multi-Agent systems.

40%

Of enterprise apps will feature AI agents by the end of 2026.

Components

Integrated Goals, Memory, Tools, and advanced Reasoning engines.

Ethics & Safety

Governance frameworks, bias mitigation, and responsible AI transparency.

\$450B

Projected market value of autonomous AI systems by 2028.



Hands-on Project: Build an autonomous task assistant that plans and executes independently.

MODULE 8: AI AGENT DEVELOPMENT

Building Individual Autonomous AI Agents

LangChain

Modular building blocks for LLM-powered applications.

CrewAI

Orchestrating autonomous multi-agent crews for complex tasks.

AutoGen

Enabling conversational agents that can talk to each other.

LlamaIndex

The premier data framework for LLM applications.

Tool Integrations

Connect agents to APIs, databases, and external services for real-world action.

Agent Memory

Implement long-term memory and context management for consistent agent behavior.

Workflow Automation

Design and deploy agents that automate complex, multi-step business workflows.



Hands-on Project: Develop an AI workflow automation agent from scratch.

MODULE 9: MULTI-AGENT SYSTEMS

Orchestrating and Scaling Collaborative AI Agents

Task Orchestration

Master planning workflows, task delegation, and collaborative system design.

Agent Communication

Implement message passing protocols and cross-agent coordination strategies.

Debugging & Scaling

Ensure observability and performance monitoring for complex multi-agent architectures.

38%

Of companies will treat AI agents as team members by 2028.

 **Hands-on Project:** Build a comprehensive multi-agent AI automation system.

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Master the future of technology and transform your career.

ENROLL NOW

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